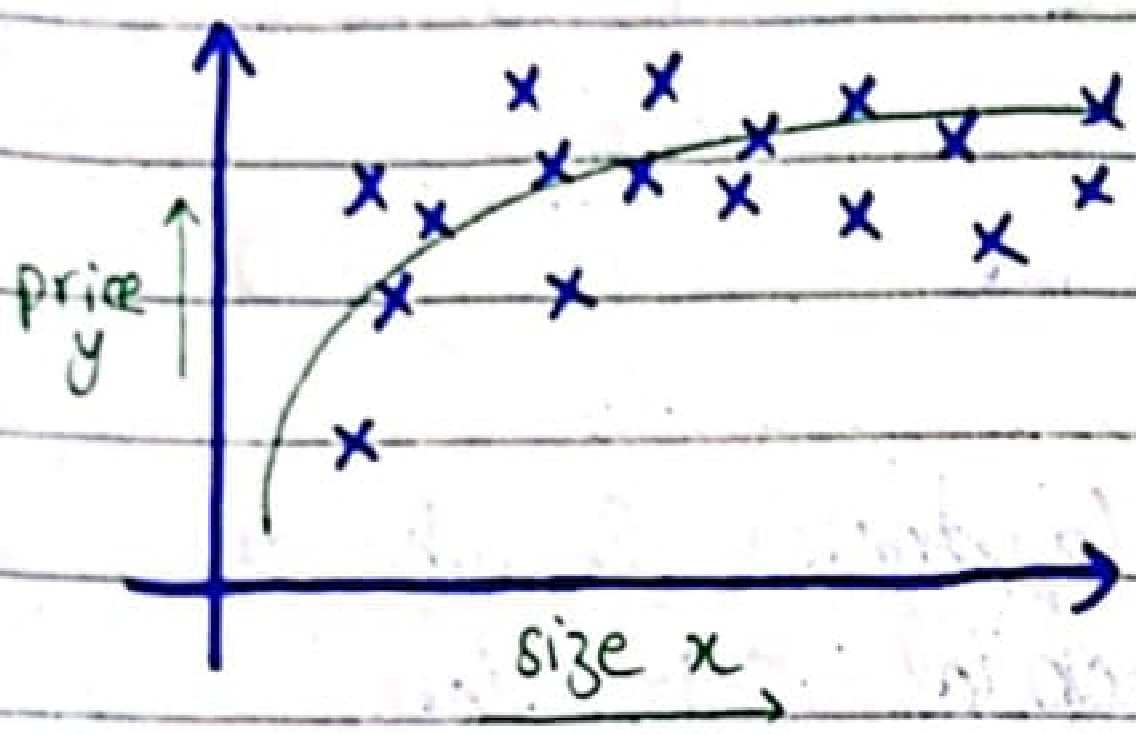
Polynomial Regression

Let's use the idea of multi-linear regression & feature engineering to come up with a new algorithm called polynomial regression.
it'll let you fit curves, not linear functions to your data.

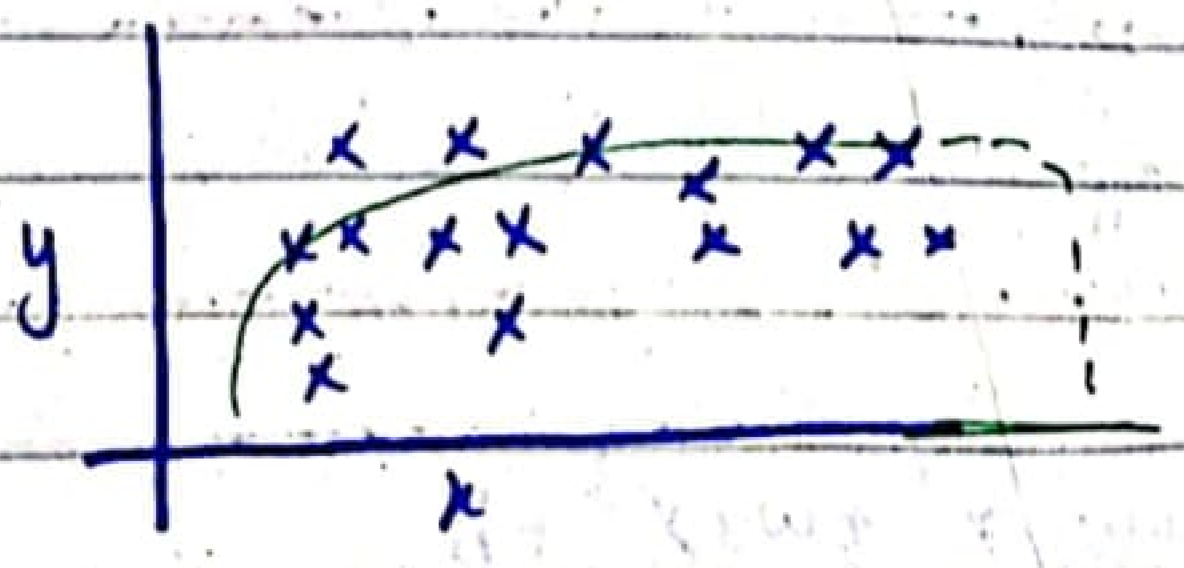


housing data set.
(doesn't look that straight fit this data well).

Maybe you want to fit a curve, maybe a quadratic function to data like:

$$f_{\vec{w},b}(x) = \underset{\text{size}}{w_1 x} + \underset{\text{size}^2}{w_2 x^2} + b \longrightarrow$$ maybe give better fit to data.

but then you may decide this quad model doesn't really make sense bcz quadratic function eventually comes back down,

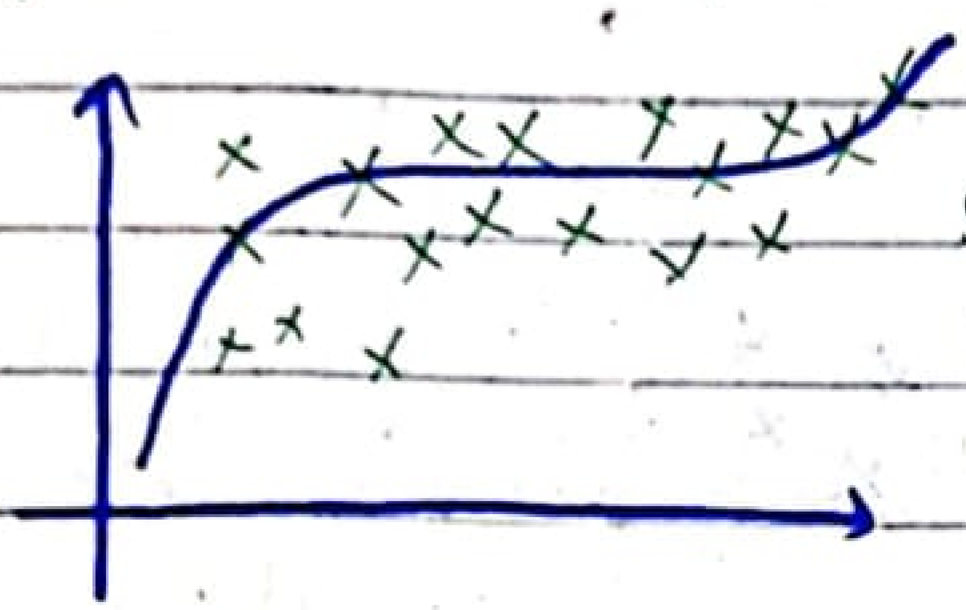


we wouldn't really expect prices to go down if size increases though.

So then, you may choose a cubic function:

$$f_{\vec{w},b}(\vec{x}) = w_1 x + w_2 x^2 + w_3 x^3 + b$$

maybe this model produces this curve,



somewhat better fit to data bcoz size come eventually back up.

These two square (quadratic) & cubic are examples of **polynomial regression**.

Pointing out one thing:
if you create features that are these powers like square of og features like this then, **feature scaling** becomes increasingly imp.

as if:
size of house ranges from "1-1000". then second feature will range from "one to million" & third will range from "one to billion".

$\cancel{f(\vec{w},b)}$

$$f(\vec{x}) = w_1 \underset{1\times10^3}{x} + w_2 \underset{1\times10^6}{x^2} + w_3 \underset{1\times10^9}{x^3} + b$$